

# Product Carbon Footprint Study Report

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Product Name: hard disk drive production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

## I. General Information

**Producer Name:** [Information provided by the user]

**Address:** [Information provided by the user]

**Legal Representative:** [Information provided by the user]

**Product Name:** hard disk drive production

**Product Function:** [Information provided by the user]

**Based on the Standard:** ecoinvent - EF v3.1

## II. Quantitative Purpose

1 item hard disk drive production, for desktop computer

## III. Scope of Quantification

### 1. Functional or Declaratory Units

Functional or declarative units in 1 item hard disk drive production, for desktop computer.

### 2. System Boundary

hard disk drive production, for desktop computer, in China, in 2025

### 3. Cut-off Criteria

[Information provided by the user]

### 4. Time Scale

[Information provided by the user]

## IV. Inventory Analysis

### 1. Description of Data Sources

**Primary Data:** [Information provided by the user]

**Secondary Data:** ecoinvent

### 2. Principles and Procedures for Allocation

**Basis of Allocation:** [Information provided by the user]

**Allocation Procedure:** [Information provided by the user]

**Specific Allocations:** [Information provided by the user]

### 3. Data Quality Evaluation (Optional)

[Information provided by the user]

## V. Impact Assessment

### 1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

### 2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of hard disk drive production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **14.739 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

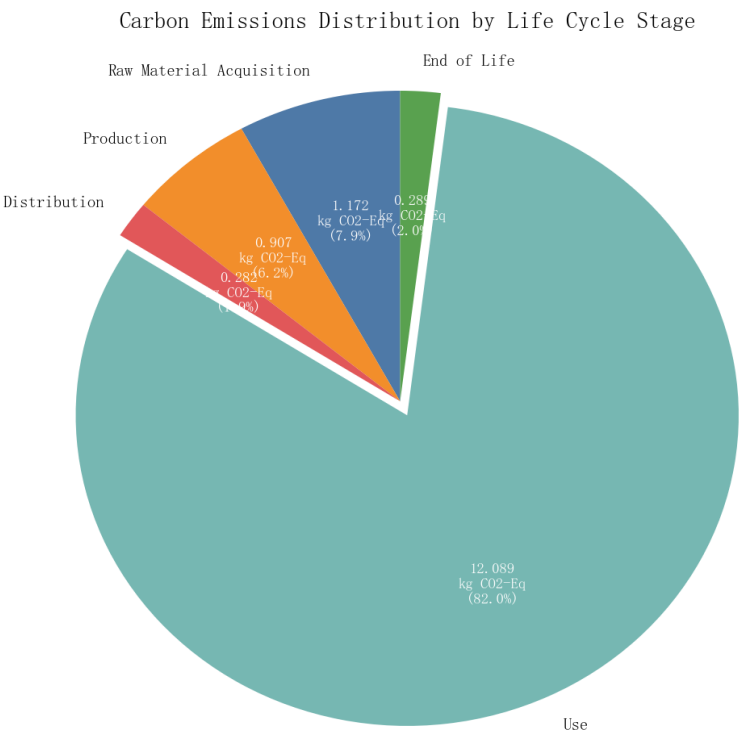


Table 1: Carbon Emissions by Life Cycle Stage

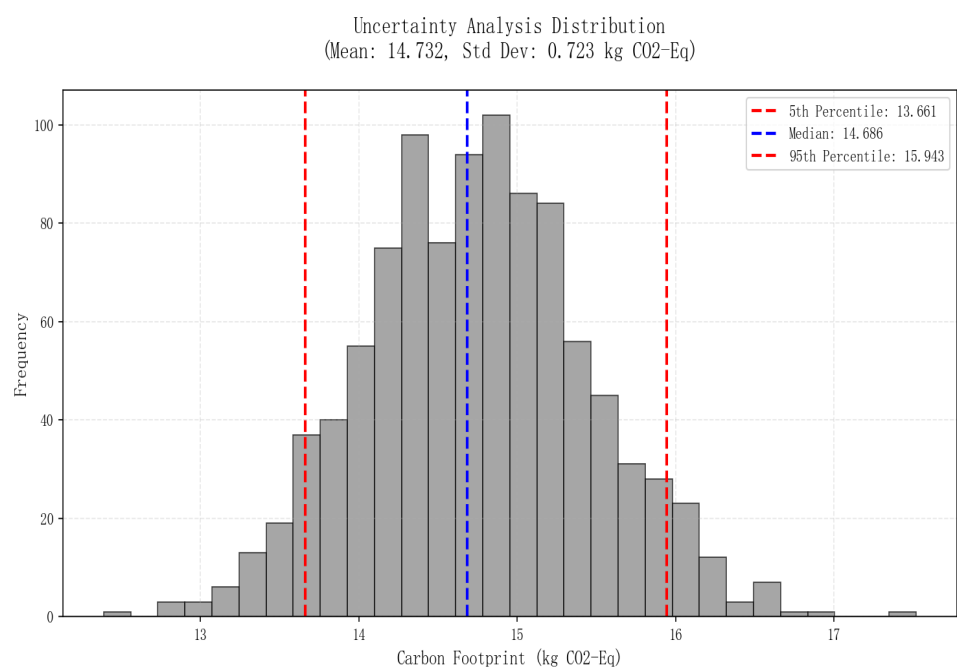
| Life Cycle Stage         | Carbon Footprint (kg CO2-Eq) | Percentage (%) |
|--------------------------|------------------------------|----------------|
| Raw Material Acquisition | 1.172                        | 7.95           |
| Production               | 0.907                        | 6.15           |
| Distribution             | 0.282                        | 1.92           |

|              |               |               |
|--------------|---------------|---------------|
| Use          | 12.089        | 82.02         |
| End of Life  | 0.289         | 1.96          |
| <b>Total</b> | <b>14.739</b> | <b>100.00</b> |

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



| Statistical Metric            | Value (kg CO2-Eq) |
|-------------------------------|-------------------|
| Mean                          | 14.732            |
| Standard Deviation            | 0.723             |
| Coefficient of Variation (CV) | 4.91%             |
| Median                        | 14.686            |
| 5th Percentile                | 13.661            |
| 95th Percentile               | 15.943            |

## VI. Interpretation of Results

### 1. Description of Results

The total carbon footprint of the hard disk drive is quantified at 14.739 kg CO<sub>2</sub>-Eq, representing a moderate impact level for electronic storage devices when assessed from a cradle-to-grave perspective. This aggregate value integrates emissions across all lifecycle stages, with the use phase emerging as the overwhelmingly dominant contributor at 82.0% (12.089 kg CO<sub>2</sub>-Eq). Such a distribution is typical for data storage products, where prolonged operational electricity consumption in computing environments drives the majority of greenhouse gas releases.

Analysis of stage contributions reveals that raw material acquisition (7.9%) and production (6.2%) together account for 14.1% of the total, while distribution (1.9%) and end-of-life treatment (2.0%) remain marginal. This pattern underscores the energy-intensive nature of HDD manufacturing inputs alongside the extended service life emissions. The primary emission source is unequivocally the use stage, reflecting electricity demand during active read/write operations and idle periods, which far exceeds impacts from upstream material processing or downstream logistics.

### 2. Statement of Assumptions and Limitations

Key assumptions in this study include an average product lifespan of five years under standard office or data center usage patterns, with electricity modeled using a global grid mix and typical HDD power consumption rates of 6-8 watts during operation. Additional assumptions encompass average transportation distances for distribution and standard recycling rates for end-of-life processing based on European waste management practices.

Limitations of data sources and quality arise from reliance on secondary databases such as Ecoinvent for background processes, which may not fully capture site-specific manufacturing variations or regional energy mixes. Primary data gaps for certain alloy compositions and semiconductor fabrication could introduce uncertainties of up to 15-20% in upstream impacts. System boundary restrictions exclude indirect emissions from infrastructure maintenance, software-related energy use, and potential rebound effects from increased data storage demand, limiting the assessment to direct product-related flows from raw material extraction through final disposal.

### 3. Suggestions for Improvement

- Target the use phase by developing energy-efficient HDD models with lower idle power draw and promoting renewable energy sourcing for data centers to cut the 82% operational emissions.
- Implement closed-loop material recovery for rare earth magnets and aluminum components in raw material and end-of-life stages to reduce virgin resource dependency and associated 7.9% footprint.
- Optimize production processes through renewable-powered manufacturing facilities and lean energy management to lower the 6.2% production contribution.

- Enhance data quality by collecting primary site-specific energy and material data from suppliers and conducting sensitivity analyses on usage scenarios.
- Explore lightweight packaging redesigns and localized distribution networks to further minimize the already low 1.9% transport emissions.